

# NUDGE

## Improving Decisions About Health, Wealth, and Happiness

Richard H. Thaler  
Cass R. Sunstein

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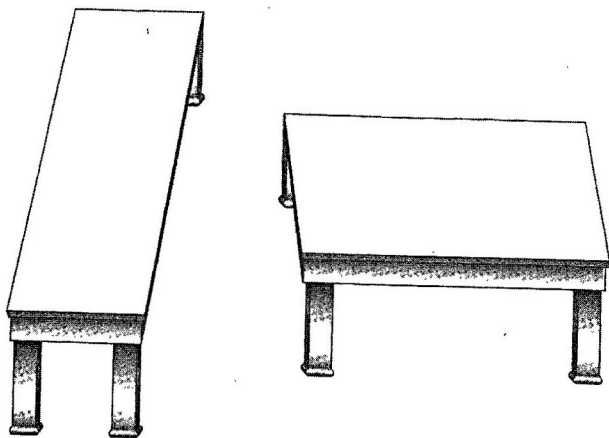
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# 1

## BIASES AND BLUNDERS

Have a look, if you will, at these two tables:

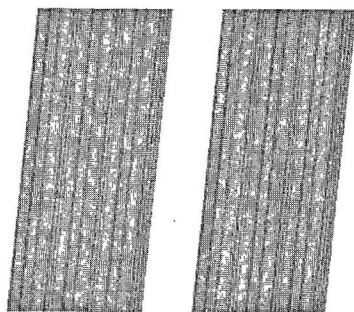


I.I.

Two tables (Adapted from Shepard [1990])

Suppose that you are thinking about which one would work better as a coffee table in your living room. What would you say are the dimensions of the two tables? Take a guess at the ratio of the length to the width of each. Just eyeball it.

If you are like most people, you think that the table on the left is much longer and narrower than the one on the right. Typical guesses are that the ratio of the length to the width is 3:1 for the left table and 1.5:1 for the right



1.2.

Tabletops (Adapted from Shepard [1990])

table. Now take out a ruler and measure each table. You will find that the two table tops are identical. Measure them until you are convinced, because this is a case where seeing is not believing. (When Thaler showed this example to Sunstein at their usual lunch haunt, Sunstein grabbed his chopstick to check.)

What should we conclude from this example? If you see the left table as longer and thinner than the right one, you are certifiably human. There is nothing wrong with you (well, at least not that we can detect from this test). Still, your judgment in this task was biased, and predictably so. No one thinks that the right table is thinner! Not only were you wrong; you were probably confident that you were right. If you like, you can put this visual to good use when you encounter others who are equally human and who are disposed to gamble away their money, say, at a bar.

Now consider Figure 1.2. Do these two shapes look the same or different? Again, if you are human, and have decent vision, you probably see these shapes as being identical, as they are. But these two shapes are just the table tops from Figure 1.1, removed from their legs and reoriented. Both the legs and the orientation facilitate the illusion that the table tops are different in Figure 1.1, so removing these distracters restores the visual system to its usual amazingly accurate state.\*

\*One of the tricks used in drawing these tables is that vertical lines look longer than horizontal lines. As a result, the Gateway Arch in St. Louis looks taller than it is wide, although the height actually equals the width.

These two figures capture the key insight that behavioral economists have borrowed from psychologists. Normally the human mind works remarkably well. We can recognize people we have not seen in years, understand the complexities of our native language, and run down a flight of stairs without falling. Some of us can speak twelve languages, improve the fanciest computers, and/or create the theory of relativity. However, even Einstein would probably be fooled by those tables. That does not mean something is wrong with us as humans, but it does mean that our understanding of human behavior can be improved by appreciating how people systematically go wrong.

To obtain that understanding, we need to explore some aspects of human thinking. Knowing something about the visual system allowed Roger Shepard (1990), a psychologist and artist, to draw those deceptive tables. He knew what to draw to lead our mind astray. Knowing something about the cognitive system has allowed others to discover systematic biases in the way we think.

## How We Think: Two Systems



The workings of the human brain are more than a bit befuddling. How can we be so ingenious at some tasks and so clueless at others? Beethoven wrote his incredible ninth symphony while he was deaf, but we would not be at all surprised if we learned that he often misplaced his house keys. How can people be simultaneously so smart and so dumb? Many psychologists and neuroscientists have been converging on a description of the brain's functioning that helps us make sense of these seeming contradictions. The approach involves a distinction between two kinds of thinking, one that is intuitive and automatic, and another that is reflective and rational.<sup>1</sup> We will call the first the Automatic System and the second the Reflective System. (In the psychology literature, these two systems are sometimes referred to as System 1 and System 2, respectively.) The key features of each system are shown in Table 1.1.



The Automatic System is rapid and is or feels instinctive, and it does not involve what we usually associate with the word *thinking*. When you duck because a ball is thrown at you unexpectedly, or get nervous when your airplane hits turbulence, or smile when you see a cute puppy, you are using



Table 1.1  
Two cognitive systems

| <i>Automatic System</i> | <i>Reflective System</i> |
|-------------------------|--------------------------|
| Uncontrolled            | Controlled               |
| Effortless              | Effortful                |
| Associative             | Deductive                |
| Fast                    | Slow                     |
| Unconscious             | Self-aware               |
| Skilled                 | Rule-following           |

your Automatic System. Brain scientists are able to say that the activities of the Automatic System are associated with the oldest parts of the brain, the parts we share with lizards (as well as puppies).<sup>2</sup>

The Reflective System is more deliberate and self-conscious. We use the Reflective System when we are asked, “How much is 411 times 37?” Most people are also likely to use the Reflective System when deciding which route to take for a trip and whether to go to law school or business school. When we are writing this book we are (mostly) using our Reflective Systems, but sometimes ideas pop into our heads when we are in the shower or taking a walk and not thinking at all about the book, and these probably are coming from our Automatic Systems. (Voters, by the way, seem to rely primarily on their Automatic System.<sup>3</sup> A candidate who makes a bad first impression, or who tries to win votes by complex arguments and statistical demonstrations, may well run into trouble.)\*

Most Americans have an Automatic System reaction to a temperature given in Fahrenheit but have to use their Reflective System to process a temperature given in Celsius; for Europeans, the opposite is true. People speak their native languages using their Automatic Systems and tend to struggle to speak another language using their Reflective Systems. Being truly bilingual means that you speak two languages using the Automatic System. Accomplished chess players and professional athletes have pretty

\*It is possible to predict the outcome of congressional elections with frightening accuracy simply by asking people to look quickly at pictures of the candidates and say which one looks more competent. These judgments, by students who did not know the candidates, forecast the winner of the election two-thirds of the time! (Toderov et al. [2005]; Benjamin and Shapiro [2007])

fancy intuitions; their Automatic Systems allow them to size up complex situations rapidly and to respond with both amazing accuracy and exceptional speed.

One way to think about all this is that the Automatic System is your gut reaction and the Reflective System is your conscious thought. Gut feelings can be quite accurate, but we often make mistakes because we rely too much on our Automatic System. The Automatic System says that “the airplane is shaking, I’m going to die,” while the Reflective System responds, “Planes are very safe!” The Automatic System says, “That big dog is going to hurt me,” and the Reflective System replies, “Most pets are quite sweet.” (In both cases, the Automatic System is squawking all the time.) The Automatic System starts out with no idea how to play golf or tennis. Note, however, that countless hours of practice enable an accomplished golfer to avoid reflection and to rely on her Automatic System—so much so that good golfers, like other good athletes, know the hazards of “thinking too much” and might well do better to “trust the gut,” or “just do it.” The Automatic System can be trained with lots of repetition—but such training takes a lot of time and effort. One reason why teenagers are such risky drivers is that their Automatic Systems have not had much practice, and using the Reflective System is much slower.

To see how intuitive thinking works, try the following little test. For each of the three questions, begin by writing down the first answer that comes to your mind. Then pause to reflect.

1. A bat and ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. How much does the ball cost? \_\_\_\_\_ cents
2. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets? \_\_\_\_\_ minutes
3. In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake? \_\_\_\_\_ days

What were your initial answers? Most people say 10 cents, 100 minutes, and 24 days. But all these answers are wrong. If you think for a minute, you will see why. If the ball costs 10 cents and the bat costs one dollar more than the ball, meaning \$1.10, then together they cost \$1.20, not \$1.10. No one who bothers to check whether his initial answer of 10 cents could pos-

sibly be right would give that as an answer, but research by Shane Frederick (2005) (who calls this series of questions the cognitive reflection test) finds that these are the most popular answers even among bright college students.

The correct answers are 5 cents, 5 minutes, and 47 days, but you knew that, or at least your Reflective System did if you bothered to consult it. Econs never make an important decision without checking with their Reflective Systems (if they have time). But Humans sometimes go with the answer the lizard inside is giving without pausing to think. If you are a television fan, think of Mr. Spock of *Star Trek* fame as someone whose Reflective System is always in control. (Captain Kirk: "You'd make a splendid computer, Mr. Spock." Mr. Spock: "That is very kind of you, Captain!") In contrast, Homer Simpson seems to have forgotten where he put his Reflective System. (In a commentary on gun control, Homer once replied to a gun store clerk who informed him of a mandatory five-day waiting period before buying a weapon, "Five days? But I'm mad now!")

One of our major goals in this book is to see how the world might be made easier, or safer, for the Homers among us (and the Homer lurking somewhere in each of us). If people can rely on their Automatic Systems without getting into terrible trouble, their lives should be easier, better, and longer.

## Rules of Thumb

Most of us are busy, our lives are complicated, and we can't spend all our time thinking and analyzing everything. When we have to make judgments, such as guessing Angelina Jolie's age or the distance between Cleveland and Philadelphia, we use simple rules of thumb to help us. We use rules of thumb because most of the time they are quick and useful.

In fact, there is a great collection edited by Tom Parker titled *Rules of Thumb*. Parker wrote the book by asking friends to send him good rules of thumb. For example, "One ostrich egg will serve 24 people for brunch." "Ten people will raise the temperature of an average size room by one degree per hour." And one to which we will return: "No more than 25 percent of the guests at a university dinner party can come from the economics department without spoiling the conversation."

Although rules of thumb can be very helpful, their use can also lead to systematic biases. This insight, first developed decades ago by two Israeli psychologists, Amos Tversky and Daniel Kahneman (1974), has changed the way psychologists (and eventually economists) think about thinking. Their original work identified three heuristics, or rules of thumb—anchoring, availability, and representativeness—and the biases that are associated with each. Their research program has come to be known as the “heuristics and biases” approach to the study of human judgment. More recently, psychologists have come to understand that these heuristics and biases emerge from the interplay between the Automatic System and the Reflective System. Let’s see how.

### *Anchoring*

Suppose we are asked to guess the population of Milwaukee, a city about two hours north of Chicago, where we live. Neither of us knows much about Milwaukee, but we think that it is the biggest city in Wisconsin. How should we go about guessing? Well, one thing we could do is start with something we do know, which is the population of Chicago, roughly three million. So we might think, Milwaukee is a major city, but clearly not as big as Chicago, so, hmmm, maybe it is one-third the size, say one million. Now consider someone from Green Bay, Wisconsin, who is asked the same question. She also doesn’t know the answer, but she does know that Green Bay has about one hundred thousand people and knows that Milwaukee is larger, so guesses, say, three times larger—three hundred thousand.

This process is called “anchoring and adjustment.” You start with some anchor, the number you know, and adjust in the direction you think is appropriate. So far, so good. The bias occurs because the adjustments are typically insufficient. Experiments repeatedly show that, in problems similar to our example, people from Chicago are likely to make a high guess (based on their high anchor) while those from Green Bay guess low (based on their low anchor). As it happens, Milwaukee has about 580,000 people.<sup>4</sup>

Even obviously irrelevant anchors creep into the decision-making process. Try this one yourself. Take the last three digits of your phone number and add two hundred. Write the number down. Now, when do you think Attila the Hun sacked Europe? Was it before or after that year? What is your best guess? (We will give you one hint: It was after the birth of Jesus.) Even

if you do not know much about European history, you do know enough to know that whenever Attila did whatever he did, the date has nothing to do with your phone number. Still, when we conduct this experiment with our students, we get answers that are more than three hundred years later from students who start with high anchors rather than low ones. (The right answer is 411.)

Anchors can even influence how you think your life is going. In one experiment, college students were asked two questions: (a) How happy are you? (b) How often are you dating? When the two questions were asked in this order the correlation between the two questions was quite low (.11). But when the question order was reversed, so that the dating question was asked first, the correlation jumped to .62. Apparently, when prompted by the dating question, the students use what might be called the “dating heuristic” to answer the question about how happy they are. “Gee, I can’t remember when I last had a date! I must be miserable.” Similar results can be obtained from married couples if the dating question is replaced by a lovemaking question.<sup>5</sup>

In the language of this book, anchors serve as nudges. We can influence the figure you will choose in a particular situation by ever-so-subtly suggesting a starting point for your thought process. When charities ask you for a donation, they typically offer you a range of options such as \$100, \$250, \$1,000, \$5,000, or “other.” If the charity’s fund-raisers have an idea of what they are doing, these values are not picked at random, because the options influence the amount of money people decide to donate. People will give more if the options are \$100, \$250, \$1,000, and \$5,000, than if the options are \$50, \$75, \$100, and \$150.

In many domains, the evidence shows that, within reason, the more you ask for, the more you tend to get. Lawyers who sue cigarette companies often win astronomical amounts, in part because they have successfully induced juries to anchor on multimillion-dollar figures. Clever negotiators often get amazing deals for their clients by producing an opening offer that makes their adversary thrilled to pay half that very high amount.

### *Availability*

How much should you worry about hurricanes, nuclear power, terrorism, mad cow disease, alligator attacks, or avian flu? And how much

care should you take in avoiding risks associated with each? What, exactly, should you do to prevent the kinds of dangers that you face in ordinary life?

In answering questions of this kind, most people use what is called the availability heuristic. They assess the likelihood of risks by asking how readily examples come to mind. If people can easily think of relevant examples, they are far more likely to be frightened and concerned than if they cannot. A risk that is familiar, like that associated with terrorism in the aftermath of 9/11, will be seen as more serious than a risk that is less familiar, like that associated with sunbathing or hotter summers. Homicides are more available than suicides, and so people tend to believe, wrongly, that more people die from homicide.

Accessibility and salience are closely related to availability, and they are important as well. If you have personally experienced a serious earthquake, you're more likely to believe that an earthquake is likely than if you read about it in a weekly magazine. Thus vivid and easily imagined causes of death (for example, tornadoes) often receive inflated estimates of probability, and less-vivid causes (for example, asthma attacks) receive low estimates, even if they occur with a far greater frequency (here a factor of twenty). So, too, recent events have a greater impact on our behavior, and on our fears, than earlier ones. In all these highly available examples, the Automatic System is keenly aware of the risk (perhaps too keenly), without having to resort to any tables of boring statistics.

The availability heuristic helps to explain much risk-related behavior, including both public and private decisions to take precautions. Whether people buy insurance for natural disasters is greatly affected by recent experiences.<sup>6</sup> In the aftermath of an earthquake, purchases of new earthquake insurance policies rise sharply—but purchases decline steadily from that point, as vivid memories recede. If floods have not occurred in the immediate past, people who live on floodplains are far less likely to purchase insurance. And people who know someone who has experienced a flood are more likely to buy flood insurance for themselves, regardless of the flood risk they actually face.

Biased assessments of risk can perversely influence how we prepare for and respond to crises, business choices, and the political process. When Internet stocks have done very well, people might well buy Internet stocks,

even if by that point they've become a bad investment. Or suppose that people falsely think that some risks (a nuclear power accident) are high, whereas others (a stroke) are relatively low. Such misperceptions can affect policy, because governments are likely to allocate their resources in a way that fits with people's fears rather than in response to the most likely danger.

When "availability bias" is at work, both private and public decisions may be improved if judgments can be nudged back in the direction of true probabilities. A good way to increase people's fear of a bad outcome is to remind them of a related incident in which things went wrong; a good way to increase people's confidence is to remind them of a similar situation in which everything worked out for the best. The pervasive problems are that easily remembered events may inflate people's probability judgments, and that if no such events come to mind, their judgments of likelihoods might be distorted downward.

### *Representativeness*

The third of the original three heuristics bears an unwieldy name: representativeness. Think of it as the similarity heuristic. The idea is that when asked to judge how likely it is that A belongs to category B, people (and especially their Automatic Systems) answer by asking themselves how similar A is to their image or stereotype of B (that is, how "representative" A is of B). Like the other two heuristics we have discussed, this one is used because it often works. We think a 6-foot-8-inch African-American man is more likely to be a professional basketball player than a 5-foot-6-inch Jewish guy because there are lots of tall black basketball players and not many short Jewish ones (at least not these days). Stereotypes are sometimes right!

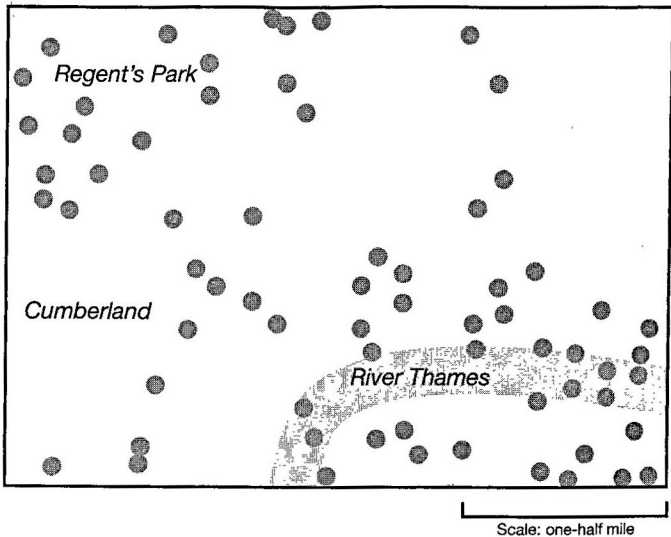
Again, biases can creep in when similarity and frequency diverge. The most famous demonstration of such biases involves the case of a hypothetical woman named Linda. In this experiment, subjects were told the following: "Linda is thirty-one years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice and also participated in antinuclear demonstrations." Then people were asked to rank, in order of the probability of their occurrence, eight possible futures for Linda. The two crucial

answers were “bank teller” and “bank teller and active in the feminist movement.” Most people said that Linda was less likely to be a bank teller than to be a bank teller and active in the feminist movement.

This is an obvious logical mistake. It is, of course, not logically possible for any two events to be more likely than one of them alone. It just has to be the case that Linda is more likely to be a bank teller than a feminist bank teller, because all feminist bank tellers are bank tellers. The error stems from the use of the representativeness heuristic: Linda’s description seems to match “bank teller and active in the feminist movement” far better than “bank teller.” As Stephen Jay Gould (1991) once observed, “I know [the right answer], yet a little homunculus in my head continues to jump up and down, shouting at me—‘but she can’t just be a bank teller; read the description!’” Gould’s homunculus is the Automatic System in action.

Use of the representativeness heuristic can cause serious misperceptions of patterns in everyday life. When events are determined by chance, such as a sequence of coin tosses, people expect the resulting string of heads and tails to be representative of what they think of as random. Unfortunately, people do not have accurate perceptions of what random sequences look like. When they see the outcomes of random processes, they often detect patterns that they think have great meaning but in fact are just due to chance. You might flip a coin three times, see it come up heads every time, and conclude that there is something funny about the coin. But the fact is that if you flip any coin a lot, it won’t be so unusual to see three heads in a row. (Try it and you’ll see. As a little test, Sunstein, having just finished this paragraph, flipped a regular penny three times—and got heads every time. He was amazed. He shouldn’t have been.)

A less trivial example, from the Cornell psychologist Tom Gilovich (1991), comes from the experience of London residents during the German bombing campaigns of World War II. London newspapers published maps, such as the one shown in Figure 1.3, displaying the location of the strikes from German V-1 and V-2 missiles that landed in central London. As you can see, the pattern does not seem at all random. Bombs appear to be clustered around the River Thames and also in the northwest sector of the map. People in London expressed concern at the time because the pattern seemed to suggest that the Germans could aim their bombs with great precision. Some Londoners even speculated that the blank spaces



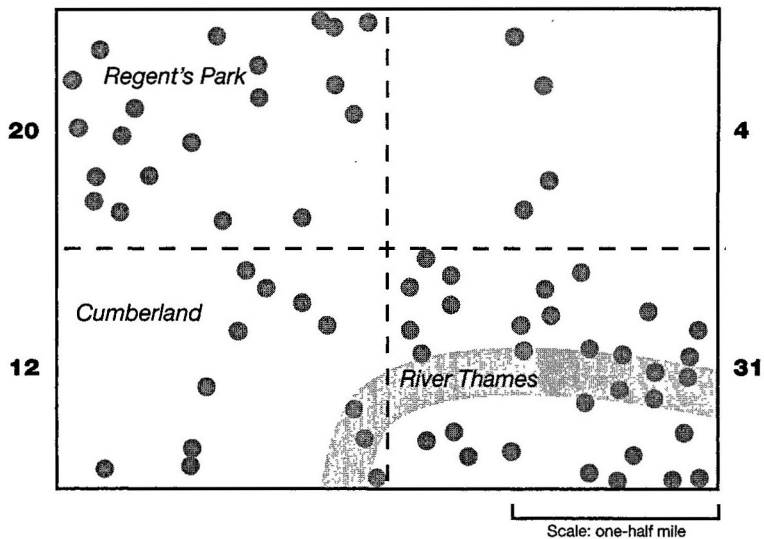
1.3.

Map of London showing V-1 rocket strikes (Adapted from Gilovich [1991])

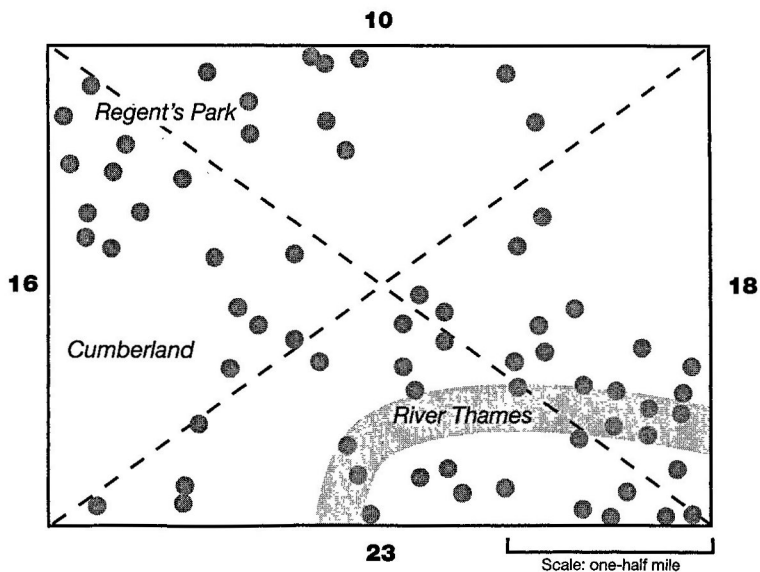
were probably the neighborhoods where German spies lived. They were wrong. In fact the Germans could do no better than aim their bombs at Central London and hope for the best. A detailed statistical analysis of the dispersion of the location of the bomb strikes determined that within London the distribution of bomb strikes was indeed random.

Still, the location of the bomb strikes does not *look* random. What is going on here? We often see patterns because we construct our informal tests only after looking at the evidence. The World War II example is an excellent illustration of this problem. Suppose we divide the map into quadrants, as in Figure 1.4a. If we then do a formal statistical test—or, for the less statistically inclined, just count the number of hits in each quadrant—we do find evidence of a nonrandom pattern. However, nothing in nature suggests that this is the right way to test for randomness. Suppose instead we form the quadrants diagonally as in Figure 1.4b. We are now unable to reject the hypothesis that the bombs land at random. Unfortunately, we do not subject our own perceptions to such rigorous alternative testing.

Gilovich (with colleagues Vallone and Tversky [1985]) is also responsible for perhaps the most famous (or infamous) example of misperception of randomness, namely the widely held view among basketball fans that



a



b

1.4.

Map of London showing V-I rocket strikes, with vertical-horizontal grid (a) and diagonal grid (b). The figures outside the grid refer to the number of dots in the quadrant. (Adapted from Gilovich [1991])

there is a strong pattern of “streak shooting.” We will not go into this in detail, because our experience tells us that the cognitive illusion here is so powerful that most people (influenced by their Automatic System) are unwilling even to consider the possibility that their strongly held beliefs might be wrong. But here is the short version. Most basketball fans think that a player is more likely to make his next shot if he has made his last shot, or even better, his last few shots. Players who have hit a few shots in a row, or even most of their recent shots, are said to have a “hot hand,” which is taken by all sports announcers to be a good signal about the future. Passing the ball to the player who is hot is taken to be an obvious bit of good strategy.

It turns out that the “hot hand” is just a myth. Players who have made their last few shots are no more likely to make their next shot (actually a bit less likely). Really.

Once people are told these facts, they quickly start forming alternative versions of the hot-hand theory. Maybe the defense adjusts and guards the “hot” player more closely. Maybe the hot player adjusts and starts taking harder shots. These are fine observations that need to be investigated. But notice that, before seeing the data, when fans were asked about actual shooting percentages after a series of made shots, they routinely subscribed to the hot-hand theory—no qualifiers were thought necessary. Many researchers have been so sure that the original Gilovich results were wrong that they set out to find the hot hand. To date, no one has found it.<sup>7</sup>

Jay Koehler and Caryn Conley (2003) performed a particularly clean test using the annual three-point shooting contest held at the National Basketball Association All-Star Game. In this contest, the players (among the best three-point shooters in the league) take a series of shots from behind the three-point shooting arc. Their goal is to make as many shots as possible in sixty seconds. Without any defense or alternative shots, this would seem to be an ideal situation in which to observe the hot hand. However, as in the original study, there was no evidence of any streakiness. This absence of streak shooting did not stop the announcers from detecting sudden temperature variations in the players. (“Dana Baros is hot!” “Legler is on fire!”) But these outbursts by the announcers had no predictive power. Before the announcers spoke of hotness, the players had made

80.5 percent of their three previous shots. After the hotness pronouncements, players made only 55.2 percent—not significantly better than their overall shooting percentage in the contest, 53.9 percent.

Of course, it is no great problem if basketball fans are confused about what they see when they are watching games on television. But the same cognitive biases occur in other, more weighty domains. Consider the phenomenon of “cancer clusters.” These can cause a great deal of private and public consternation, and they often attract sustained investigations, designed to see what on earth (or elsewhere) could possibly have caused a sudden and otherwise inexplicable outbreak of cancer cases. Suppose that in a particular neighborhood we find an apparently elevated cancer rate—maybe ten people, in a group of five hundred, have been diagnosed with cancer within the same six-month period. Maybe all ten people live within three blocks of one another. And in fact, American officials receive reports of more than one thousand suspected cancer clusters every year, with many of these suspected clusters investigated further for a possible “epidemic.”<sup>8</sup>

The problem is that in a population of three hundred million, it is inevitable that certain neighborhoods will see unusually high cancer rates within any one-year period. The resulting “cancer clusters” may be products of random fluctuations. Nonetheless, people insist that they could not possibly occur by chance. They get scared, and sometimes government wrongly intervenes on their behalf. Mostly, though, there is thankfully nothing to worry about, except for the fact that the use of the representativeness heuristic can cause people to confuse random fluctuations with causal patterns.

### Optimism and Overconfidence

Before the start of Thaler’s class in Managerial Decision Making, students fill out an anonymous survey on the course Web site. One of the questions is “In which decile do you expect to fall in the distribution of grades in this class?” Students can check the top 10 percent, the second 10 percent, and so forth. Since these are MBA students, they are presumably well aware that in any distribution, half the population will be in the top 50 percent and half in the bottom. And only 10 percent of the class can, in fact, end up in the top decile.

Nevertheless, the results of this survey reveal a high degree of unrealistic optimism about performance in the class. Typically less than 5 percent of the class expects their performance to be below the median (the 50th percentile) and more than half the class expects to perform in one of the top two deciles. Invariably, the largest group of students put themselves in the second decile. We think this is most likely explained by modesty. They really think they will end up in the top decile, but are too modest to say so.

MBA students are not the only ones overconfident about their abilities. The "above average" effect is pervasive. Ninety percent of all drivers think they are above average behind the wheel, even if they don't live in Lake Wobegon. And nearly everyone (including some who are rarely seen smiling) thinks he has an above-average sense of humor. (That is because they know what is funny!) This applies to professors, too. About 94 percent of professors at a large university were found to believe that they are better than the average professor, and there is every reason to think that such overconfidence applies to professors in general.<sup>9</sup> (Yes, we admit to this particular failing.)

People are unrealistically optimistic even when the stakes are high. About 50 percent of marriages end in divorce, and this is a statistic most people have heard. But around the time of the ceremony, almost all couples believe that there is approximately a zero percent chance that their marriage will end in divorce—even those who have already been divorced!<sup>10</sup> (Second marriage, Samuel Johnson once quipped, "is the triumph of hope over experience.") A similar point applies to entrepreneurs starting new businesses, where the failure rate is at least 50 percent. In one survey of people starting new businesses (typically small businesses, such as contracting firms, restaurants, and salons), respondents were asked two questions: (a) What do you think is the chance of success for a typical business like yours? (b) What is your chance of success? The most common answers to these questions were 50 percent and 90 percent, respectively, and many said 100 percent to the second question.<sup>11</sup>

Unrealistic optimism can explain a lot of individual risk taking, especially in the domain of risks to life and health. Asked to envision their future, students typically say that they are far less likely than their classmates to be fired from a job, to have a heart attack or get cancer, to be divorced after a few years of marriage, or to have a drinking problem. Gay men sys-

tematically underestimate the chance that they will contract AIDS, even though they know about AIDS risks in general. Older people underestimate the likelihood that they will be in a car accident or suffer major diseases. Smokers are aware of the statistical risks, and often even exaggerate them, but most believe that they are less likely to be diagnosed with lung cancer and heart disease than most nonsmokers. Lotteries are successful partly because of unrealistic optimism.<sup>12</sup>

Unrealistic optimism is a pervasive feature of human life; it characterizes most people in most social categories. When they overestimate their personal immunity from harm, people may fail to take sensible preventive steps. If people are running risks because of unrealistic optimism, they might be able to benefit from a nudge. In fact, we have already mentioned one possibility: if people are reminded of a bad event, they may not continue to be so optimistic.

### Gains and Losses

People hate losses (and their Automatic Systems can get pretty emotional about them). Roughly speaking, losing something makes you twice as miserable as gaining the same thing makes you happy. In more technical language, people are “loss averse.” How do we know this?

Consider a simple experiment.<sup>13</sup> Half the students in a class are given coffee mugs with the insignia of their home university embossed on it. The students who do not get a mug are asked to examine their neighbor’s mugs. Then mug owners are invited to sell their mugs and nonowners are invited to buy them. They do so by answering the question “At each of the following prices, indicate whether you would be willing to (give up your mug/buy a mug).” The results show that those with mugs demand roughly twice as much to give up their mugs as others are willing to pay to get one. Thousands of mugs have been used in dozens of replications of this experiment, but the results are nearly always the same. Once I have a mug, I don’t want to give it up. But if I don’t have one, I don’t feel an urgent need to buy one. What this means is that people do not assign specific values to objects. When they have to give something up, they are hurt more than they are pleased if they acquire the very same thing.

It is also possible to measure loss aversion with gambles. Suppose I ask

you whether you want to make a bet. Heads you win \$X, tails you lose \$100. How much does X have to be for you to take the bet? For most people, the answer to this question is somewhere around \$200. This implies that the prospect of winning \$200 just offsets the prospect of losing \$100.

Loss aversion helps produce inertia, meaning a strong desire to stick with your current holdings. If you are reluctant to give up what you have because you do not want to incur losses, then you will turn down trades you might have otherwise made. In another experiment, half the students in a class received coffee mugs (of course) and half got large chocolate bars. The mugs and the chocolate cost about the same, and in pretests students were as likely to choose one as the other. Yet when offered the opportunity to switch from a mug to a candy bar or vice versa, only one in ten switched.

As we will see, loss aversion operates as a kind of cognitive nudge, pressing us not to make changes, even when changes are very much in our interests.

### Status Quo Bias

Loss aversion is not the only reason for inertia. For lots of reasons, people have a more general tendency to stick with their current situation. This phenomenon, which William Samuelson and Richard Zeckhauser (1988) have dubbed the “status quo bias,” has been demonstrated in numerous situations. Most teachers know that students tend to sit in the same seats in class, even without a seating chart. But status quo bias can occur even when the stakes are much larger, and it can get us into a lot of trouble.

For example, in retirement savings plans, such as 401(k)s, most participants pick an asset allocation and then forget about it. In one study conducted in the late 1980s, participants in TIAA-CREF, the pension plan of many college professors, the median number of changes in the asset allocation of the lifetime of a professor was, believe it or not, zero. In other words, over the course of their careers, more than half of the participants made exactly no changes to the way their contributions were being allocated. Perhaps even more telling, many married participants who were sin-

gle when they joined the plan still have their mothers listed as their beneficiaries!

Status quo bias is easily exploited. Many years ago American Express wrote Sunstein a cheerful letter telling him that he could receive, for free, three-month subscriptions to five magazines of his choice. Free subscriptions seem like a bargain, even if the magazines rarely get read, so Sunstein happily made his choices. What he didn't realize was that unless he took some action to cancel his subscription, he would continue to receive the magazines, paying for them at the normal rate. For about a decade, he has continued to subscribe to magazines that he hardly ever reads. (He keeps intending to cancel those subscriptions, but somehow never gets around to it. We hope to get around to discussing procrastination in the next chapter.)

One of the causes of status quo bias is a lack of attention. Many people adopt what we will call the “yeah, whatever” heuristic. A good illustration is the carryover effect in television viewing. Network executives spend a lot of time working on scheduling because they know that a viewer who starts the evening on NBC tends to stay there. Since remote controls have been pervasive in this country for decades, the actual “switching” costs in this context are literally one thumb press. But when one show ends and the next one comes on, a surprisingly high number of viewers (implicitly) say, “yeah, whatever” and keep watching. Nor is Sunstein the only victim of automatic renewal of magazine subscriptions. Those who are in charge of circulation know that when renewal is automatic, and when people have to make a phone call to cancel, the likelihood of renewal is much higher than it is when people have to indicate that they actually want to continue to receive the magazine.

The combination of loss aversion with mindless choosing implies that if an option is designated as the “default,” it will attract a large market share. Default options thus act as powerful nudges. In many contexts defaults have some extra nudging power because consumers may feel, rightly or wrongly, that default options come with an implicit endorsement from the default setter, be it the employer, government, or TV scheduler. For this and other reasons, setting the best possible defaults will be a theme we explore often in the course of this book.

## Framing

Suppose that you are suffering from serious heart disease and that your doctor proposes a grueling operation. You're understandably curious about the odds. The doctor says, "Of one hundred patients who have this operation, ninety are alive after five years." What will you do? If we fill in the facts in a certain way, the doctor's statement will be pretty comforting, and you'll probably have the operation.

But suppose the doctor frames his answer in a somewhat different way. Suppose that he says, "Of one hundred patients who have this operation, ten are dead after five years." If you're like most people, the doctor's statement will sound pretty alarming, and you might not have the operation. The Automatic System thinks: "A significant number of people are dead, and I might be one of them!" In numerous experiments, people react very differently to the information that "ninety of one hundred are alive" than to the information that "ten of one hundred are dead"—even though the content of the two statements is exactly the same. Even experts are subject to framing effects. When doctors are told that "ninety of one hundred are alive," they are more likely to recommend the operation than if told that "ten of one hundred are dead."<sup>14</sup>

Framing matters in many domains. When credit cards started to become popular forms of payment in the 1970s, some retail merchants wanted to charge different prices to their cash and credit card customers. (Credit card companies typically charge retailers 1 percent of each sale.) To prevent this, credit card companies adopted rules that forbade their retailers from charging different prices to cash and credit customers. However, when a bill was introduced in Congress to outlaw such rules, the credit card lobby turned its attention to language. Its preference was that if a company charged different prices to cash and credit customers, the credit price should be considered the "normal" (default) price and the cash price a discount—rather than the alternative of making the cash price the usual price and charging a surcharge to credit card customers.

The credit card companies had a good intuitive understanding of what psychologists would come to call "framing." The idea is that choices depend, in part, on the way in which problems are stated. The point matters a great deal for public policy. Energy conservation is now receiving a lot of

attention, so consider the following information campaigns: (a) If you use energy conservation methods, you will save \$350 per year; (b) If you do not use energy conservation methods, you will lose \$350 per year. It turns out that information campaign (b), framed in terms of losses, is far more effective than information campaign (a). If the government wants to encourage energy conservation, option (b) is a stronger nudge.

Framing works because people tend to be somewhat mindless, passive decision makers. Their Reflective System does not do the work that would be required to check and see whether reframing the questions would produce a different answer. One reason they don't do this is that they wouldn't know what to make of the contradiction. This implies that frames are powerful nudges, and must be selected with caution.

### So What?

Our goal in this chapter has been to offer a brief glimpse at human fallibility. The picture that emerges is one of busy people trying to cope in a complex world in which they cannot afford to think deeply about every choice they have to make. People adopt sensible rules of thumb that sometimes lead them astray. Because they are busy and have limited attention, they accept questions as posed rather than trying to determine whether their answers would vary under alternative formulations. The bottom line, from our point of view, is that people are, shall we say, nudge-able. Their choices, even in life's most important decisions, are influenced in ways that would not be anticipated in a standard economic framework. Here is one final example to illustrate.

One of the most scenic urban thoroughfares in the world is Chicago's Lake Shore Drive, which hugs the Lake Michigan coastline that is the city's eastern boundary. The drive offers stunning views of Chicago's magnificent skyline. There is one stretch of this road that puts drivers through a series of S curves. These curves are dangerous. Many drivers fail to take heed of the reduced speed limit (25 mph) and wipe out. Recently, the city has employed a new way of encouraging drivers to slow down.

At the beginning of the dangerous curve, drivers encounter a sign painted on the road warning of the lower speed limit, and then a series of white stripes painted onto the road. The stripes do not provide much if any



1.5.

Lake Shore Drive, Chicago (Courtesy of the city of Chicago)

tactile information (they are not speed bumps) but rather just send a visual signal to drivers. When the stripes first appear, they are evenly spaced, but as drivers reach the most dangerous portion of the curve, the stripes get closer together, giving the sensation that driving speed is increasing (see Figure 1.5). One's natural instinct is to slow down. When we drive on this familiar stretch of road, we find that those lines are speaking to us, gently urging us to touch the brake before the apex of the curve. We have been nudged.